How do new products perform? An investigation.

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Introduction

Releasing new products to the market is a common tactic used by manufacturers to grow their business. New products are common. Marketing consultancies estimate that a new consumer product is introduced into the United States every two minutes (Nielsen, 2019). Given the $26m average advertising investment that is reportedly spent on successful new products (Nielsen, 2014), new product development (NPD) and launch is a substantial economic activity. Not only are new products costly, they are also risky, because around 40% of new products fail (e.g., Markham and Lee, 2013; Page, 1993; Victory et al., 2021).

While the 60% success rate is an improvement than what is widely believed, a 40% chance of failure poses a significant risk. Academics aim to guide practitioners on how to reduce these risks, by investigating how certain conditions (e.g., innovativeness) relate to certain new product success outcomes (Lee and Markham, 2016). But this research area seldom provides consistent insights, likely due to measurement inconsistencies and unclear conceptualisations (Danneels and Kleinschmidt, 2001; Salnikova et al., 2019). These mixed relationships are not surprising given the success measures used will influence the conditions that are important in achieving that type of new product success (Hart, 1993). That is, the marketing activities used to increase penetration might be different to the marketing activities used to increase profit. Addressing the issues in measuring success is essential to build empirical generalisations, that can help practitioners improve the chance of introducing a successful new product. This study benchmarks how new products, specifically new line extensions, perform on key replicable and objective performance indicators. The goal of the research is to identify how common new product success is in consumer packaged goods categories, using a suite of success measures over various time horizons.

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We investigate the performance of 36,994 new line extensions (NLEs) that were introduced in four markets in the United States over a nine-year period (2005-2013). This study is among the first to document NLE success using survival duration and five key performance indications (that each capture a different perspective in success) derived from customer purchasing data. The study makes two main theoretical contributions. First, it documents the different levels of NLE success according to different measurement approaches. This approach may reduce the reliance on surveys and subjective practitioner evaluation that has been prevalent in academic research on new product success. We also show the suite of success measures converge around one year after introduction. This highlights the importance of academic research to measure performance at least to the first year after launch. Second, we show many new product success measures have a unimodal distribution with a positive skew. In other words, 'successful' NLEs are rare, most NLEs have low market shares and penetration that are below the average in its category. This result reveals the divergence in success that NLEs achieve on key indicators. This finding helps academics to select the types of statistical analyses needed in studies that investigate the various conditions in new product success. This paper will provide academics with a foundational understanding about new product success to help untangle the relationships in new product success.

The study also has substantial value to practitioners involved in NPD activities, because it provides an understanding about how NLEs perform on important marketplace success measures; as well as how these NLs vary on them over time. First, we find the oft-quoted 80% failure rate is not reached in the shorter term (i.e., only after three years). Second, over this 3-year period, most NLEs have a market share and penetration just around or below the average in their respective category. Finally, NLEs are more often bought by category buyers who have previously bought the parent brand and majority of NLEs contribute less than 10%

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to its parent brand sales. This study provides practitioners with a benchmark about what performance to expect and when for NLEs and roadmap about which performance indicators are important to track success.

Theoretical Background

New product success is a common research area in marketing, particularly studies exploring the possible conditions affecting new product success. Given the mixed relationships that are commonly reported, disentangling these relationships is a priority in innovation research (Lee and Markham, 2016). However, when it comes to *success*, there is no single definition about what is a successful new product or one best approach to measure success. The variation in how new product success is defined and measured can vary depending who (i.e., practitioners or academics) is measuring success (Griffin and Page, 1993), the data available and used to measure success, the organisation, category, type or role of the new product itself (Griffin and Page, 1996), or even the timing around when new product success is measured (Hultink and Robben, 1995).

Likely due to the lack of guidelines around measurement, practitioners use multiple measures to evaluate new product success (Booz-Allen & Hamilton Inc., 1982; Mintz and Currim, 2013). Market share is a common measure used by marketing practitioners to evaluate NPD activities (Mintz and Currim, 2013; Nijssen, 1999). However, market share is not used as often to evaluate new product success in academic research (Griffin and Page, 1993). Instead, academic research has traditionally surveyed practitioners (Markham and Lee, 2013), which has led to the widespread use of subjective and quasi-objective measures (see Richard et al., 2009), including the percentage of the firm's new products rated as successful, and the contribution of new products to company sales (Griffin and Page, 1993).

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An oft-quoted statistic is that 80% of new products fail but published research has established much higher, and surprisingly steady, rates of successful new products than this figure (e.g., Castellion and Markham, 2013; Crawford, 1979, 1987). Much of the peer-reviewed research use practitioners to identify success (subjective measurement), which uses practitioners to recall the products that were introduced over a specific period (often five years) and judge which of these products are considered 'successful' (e.g., Barczak et al., 2009; Cooper et al., 2004; Cooper and Kleinschmidt, 1995; Edgett et al., 1992; Griffin, 1997; Hultink et al., 1997; Hultink et al., 1998; Hultink et al., 2000; Markham and Lee, 2013; Page, 1993). While practitioners are helpful in capturing the nuances in new product success, subjective measurement is prone to bias (Nijssen, 1999) and does not outline the explicit criteria or indicators used to measure success. Recent studies have used observed survival in the market to evaluate the success of new launches (e.g., Salnikova et al., 2019; Victory et al., 2021) but this still does not tell us how the new products perform on other key performance indicators. This makes it important to investigate how new products perform using specific criteria, to work towards an understanding of what performance to expect from them.

The lack of clarity on how new products perform on key metrics and how they should be evaluated necessitates the need for further research. More research needs to describe how new products perform (Cooper, 1984) because without these benchmarks, it will be near impossible to establish a generalized relationship between certain conditions and new product success. To address this important issue, we document NLE success, using market survival and five other performance indicators, derived from large-scale consumer panel data. This study explores NLE success and builds an understanding about what to expect from NLEs, to

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provide a framework to improve the evaluation and selection of new product success measures. The context of the study is consumer-package goods, in the United States market.

Research questions

New product success can be measured in many ways, likely contributing to the inconclusive conditions in new product success relationships. For example, one study identified almost 80 success measures, but certain success measures are more common among practitioners than academics (Griffin and Page, 1993). Revenue-oriented measures are used by academics and practitioners alike (Griffin and Page, 1993), whereas new product failure rate and firm sales contribution are more common among academics (e.g., Barczak et al., 2009; Markham and Lee, 2013), and market share and customer acceptance measures (e.g., penetration) are commonly used in practice (e.g., Griffin and Page, 1993; Mintz and Currim, 2013; Nijssen, 1999; Page, 1993).

To inform decision making and to build a foundation around how new products perform, we document how many new products are successful using survival; as well as five performance indicators that are used in practice and academic research, and applicable across different product categories or market sectors. The measures investigated in this paper are: 1) Failure Rate, 2**)** Market Share, 3**)** Penetration, 4**)** Repeat Buyer Rate, 5**)** Bought Parent Brand Rate and 6**)** Parent Brand Sales Share (i.e., NLE Sales Contribution to the Parent Brand). A full explanation and rationale for these measures is provided in the method section below.

We answer the following two research questions:

RQ1: How many new line extensions are successful (i.e., survived in the market) ?

RQ2: How do new line extensions perform on the five key performance indicators?

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These measures are time dependent and we examine how the NLEs perform in each quarter during the first year after launch and in the first, second and third year after introduction.

Methodology

Data source and scope

This study benchmarks new product performance, using the Nielsen consumer panel database, provided by the Kilts Center for Marketing via their academic collaboration program (Kilts Centre for Marketing, 2019). New consumer products are examined because new products introduced into consumer categories are higher risk (Lee and Markham, 2016). The Nielsen consumer panel database has purchasing, pricing, product and brand information for 1.4 million products (UPCs) in over 1,000 categories, purchased by 40,000 to 60,000 households annually, in 52 United States markets (Kilts Centre for Marketing, 2019). The Kilts Centre databases are growing in popularity in extension research (e.g., Koschmann and Sheth, 2018), likely due to the rich consumer behavior information contained in the datasets.

Only one in eight products are introduced into the United States on a national scale (Hoskins, 2015). Therefore we explored NLEs in four United States regional markets, to ensure regional and national launches are have comparable metrics. The four selected cities (*New York, NY; Chicago, IL; Los Angeles, CA; Houston, TX*). are the most populated cities in the United States (United States Census Bureau, 2015). While NLE success are calculated from a regional perspective, results by region will not be presented because there were no substantial differences in how NLEs performed by market.

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The sample investigated in this study amounted to 36,994 NLEs introduced in five consumer categories (*Cookies*, *Ground and Whole Bean Coffee*, *Ready to Eat Cereal*, *Toothpaste*, *Spray Air Fresheners*). The data spans a nine-year period (2005-2013).

New product inclusion

New products, i.e., new Universal Product Code (UPC) or Stock Keeping Unit (SKU) introductions, are investigated because manufacturers are required to make a conscious decision to introduce and allocate resources to the product, in addition to retailing partners making a decision to stock and allocate space to the product in store. Given there are many different types of new products, e.g., from new-to-the-world products, to new product lines, to product modifications (Booz-Allen & Hamilton Inc., 1982), with each type varying in their strategy and prevalence, we focus on new line extensions (NLEs). NLEs in this study are new products that supplement the portfolios of established brands in a category (Reddy et al., 1994; Tauber, 1981). NLEs are investigated, to limit the variability in strategy (and their performance expectations) and because product additions and revisions are the most common introduction type in consumer categories (Booz-Allen & Hamilton Inc., 1982; Hultink et al., 2000).

Identifying NLEs was a manual process as there is no explicit indicator flag for new products in this dataset. A NLE is a UPC introduced under an established brand name, that was not bought by any category buyers in one calendar year, but that was purchased by at least one category buyer in a subsequent year. This time period is in line with past studies that use a one year period to establish new product purchase histories (e.g., Sinapuelas et al., 2015). A similar approach has been taken to identify new product introductions in recent research (e.g., Hoskins and Griffin, 2019; Victory et al., 2021).

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New product performance measurement

We investigate NLE success using failure rate (survival) and five performance indicators, including: **a)** Market Share; **b)** Penetration; **c)** Repeat Buyer Rate; **d)** Bought Parent Brand Rate, and **e)** Parent Brand Sales Share. Together, the survival and five indicators describe whether NLEs had the potential to continue to meet objectives, the NLEs' sales success **(a)**, how consumers trial the NLEs **(b, c)** and the NLEs' incrementality to the parent brand **(f, g)**.

Failure rate: Failure rate is commonly measured by using practitioners to estimate how many new products were successful over a set time period (Barczak et al., 2009; Markham and Lee, 2013). However, practitioners do not have a standard criterion about what constitutes *new product success*. This study considers failed NLEs as NLEs that cease to exist and are no longer bought in the market (Goldenberg et al., 2001; Victory et al., 2021; Wilbur and Farris, 2014). We look forward up to three years after launch and identify when the NLEs were no longer bought. We examine failure rates over time, from the first quarter after launch to the third year after launch.

Market share: While sales revenue is a key goal for brands (Hoskins and Griffin, 2019), however revenue does not provide insight on how the new product competes, relative to other new or established products in the category. Market share is commonly used in practice to evaluate new product performance (Griffin and Page, 1996; Mintz and Currim, 2013; Nijssen, 1999; Page, 1993). We investigated how NLE performance varied when examining revenue market share and unit market share. Due to the mirrored results and the similarity of the two market share measurement approaches, we present the revenue share results for simplicity.

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Penetration: An intuitive but sometimes overlooked pattern documented in marketing is that brands vary more in the number of customers they have (i.e., higher share brands have higher penetration), rather in how often they are bought (Ehrenberg et al., 1990; Graham et al., 2017). Penetration is key metric in marketing more broadly and, after market share, is the most commonly used metric by marketers to evaluate NPD activities (Mintz and Currim, 2013). For our study we examine the quarterly NLE penetration.

Repeat buyer rate: This measure calculates the percentage of NLE buyers in one time period that had bought the NLE in a previous time period (Singh et al., 2012). The inverse calculation of new buyer rate shows how many NLE buyers were repeat buyers. We use this measure to explore how NLEs recruit and retain customers.

Bought parent brand rate: NLEs are introduced into established portfolios and are likely to cannibalize (to some degree) with the products already in the portfolio. The ideal is that NLEs will attract new buyers to the parent brand, but new variant buyers are more likely to be heavier parent brand buyers (Tanusondjaja et al., 2016). We investigate how many new buyers the NLEs provide to the parent brand by examining the percentage of NLE buyers that had purchased the parent brand previously.

Parent brand sales share: While it is important to understand cannibalisation with NLEs, less than one in three marketing practitioners measure cannibalisation in their NPD activities (Mintz and Currim, 2013; Nijssen, 1999). We measure the sales share NLEs attain within their parent brand portfolios, as used in previous portfolio research (Tanusondjaja et al.,

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2018). There was no substantial difference in the revenue and unit brand sales results. For simplicity, the revenue sales share results are presented in this paper.

This study investigates NLE success and performance each quarter, up to a total three years after introduction. This period provides a broader description of success and is longer than the commonly investigated 52-week period (e.g., Gielens and Steenkamp, 2007; Hoskins and Griffin, 2019; Sinapuelas et al., 2015; Steenkamp and Gielens, 2003). The 52-week period is cited as being acceptable in describing performance due to a belief that consumer acceptance occurs in the first six months to one year after launch (e.g., Gielens and Steenkamp, 2007; Sinapuelas et al., 2015). In examining NLE success over a longer period, we establish whether the commonly used 52-week time frame is a satisfactory timeframe to measure success or whether longer periods after the initial from launch sheds more insight.

All measures are evaluated at each quarter, including launch. One study exploring new brand extension purchasing presents results from the launch quarter but excludes this in analysis due to concerns distribution and brand awareness would not have been built in this time (Singh et al., 2012). We acknowledge all new introductions, including NLEs, will likely struggle to build distribution and awareness at launch. While success is measured in all quarters and presented in the tables, this paper will comment primarily on the results in the key yearly periods and in the quarter directly after launch (i.e., Q1) to allow for the lack of 'build up' and to exclude seasonal NLEs. Distributions for all categories were investigated but an example from one category for each key performance indicator are presented here.

Research method

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This research seeks to identify if there exists generalisable patterns for new product success. Empirical generalisations are law-like relationships where repeatable patterns are observed across conditions (Barwise, 1995; Bass, 1995; Ehrenberg, 1995; Uncles and Wright, 2004). This study uses Many Sets of Data (MSoD) to describe the patterns in NLE success indicators. Pattern spotting descriptive analyses are used, rather than significance testing, to identify 'significant sameness' of each category (Bound and Ehrenberg, 1989). The mean results for the 'typical' NLE introduced into each category are investigated and how NLEs are distributed are examined to determine how representative the 'typical' NLEs are to the majority of NLEs introduced into the market.

Results

Failure rate

Across the five categories and almost 37,000 NLEs, the sharpest quarterly increase in the NLE fail rate is in the quarter directly after launch (M= 32%, SD= 5). Half are removed from the market in the year after launch (M= 50%, SD= 6). As shown in Table 1, the NLE failure rate continues to increase in each quarter to 64% after two years (SD= 6) and reaches the famed 80% failure rate three years on (SD= 6). NLE failure rate has the largest increase in the quarter directly after launch. Failure rates then slow but continue to increase at around 4pp for each quarter.

<INSERT TABLE 1 AROUND HERE>

The highest percentage of NLEs fail in the quarter after introduction and it is likely many of these 'failures' were NLEs that were not intended to be in the market for a longer period (e.g., temporary or seasonal NLEs). The Nielsen consumer panel database does not have an

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indicator that identifies seasonal products. Once the NLEs that did not make it past the launch quarter were removed from the calculation, the failure rate decreases to around one in four NLEs in the year after launch (M= 23%, SD= 5), around half two years on (M= 47%, SD= 7) and 70% (SD= 7) three years after launch. The revised failure rates are also shown in Table 1.

The next tables present the results in each key quarter after launch, to simplify the results.

Market share

Table 2 presents the mean market share results achieved by the 'average' NLE in each category. The 'average' NLE has a market share around 0.3% (SD= 0.2%) in the quarter after launch, which continues in the first, second and third year after introduction (SD= 0.2). While the average market share results are around 0.3% in each quarter, they do not show if this is representative of what most NLE's typically achieve.

<INSERT TABLE 2 AROUND HERE>

The distribution of each success measure was investigated to understand how representative the average market share results are of NLEs. The NLE market share distributions in all categories are unimodal and positively skewed. Figure 1 is an example of how NLE market share is distributed in the Toothpaste category. The distributions show the majority of NLEs have a market share that nears zero but there is a long tail. Across the five categories, around 80% of NLEs have a market share below 0.5% in the quarter directly after introduction (M= 83%, SD= 13). A similar proportion with under 0.5% continues in the first, second and third year after launch.

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Interestingly, when comparing the market shares the NLEs have relative to the average market share in its category, 72% of NLEs across the categories had a market share below the category average in the quarter after launch (SD= 3). This decreased slightly to around two in three NLEs with a market share below the average, three years after launch (M= 66%, SD= 8). This is also shown in Table 2.

<INSERT FIGURE 1 AROUND HERE>

Penetration

Table 3 shows the 'average' NLE across the categories investigated has a penetration of 0.5% (SD= 0.2) in the quarter after launch. One year on, NLEs have a mean penetration around 0.6% (SD= 0.2). The 0.6% share continues to persist in the second and third year after launch (SD= 0.3). The stability in penetration for the 'average' NLE, particularly after the first year, provides evidence that the first 52-weeks after launch is important to customer acceptance.

<INSERT TABLE 3 AROUND HERE>

The mean penetration results are not representative of what the 'typical' NLE achieves. The penetration NLEs achieve have a unimodal distribution with a positive skew, in each period after introduction (see Cookies example in Figure 2). Across the five categories, most NLEs have a penetration of less than 0.5% in the quarter after launch (M= 72%, SD= 15) and this continues in each quarter up to three years after launch (M= 68%, SD= 17). This pattern is demonstrated in Figure 2, albeit higher in Cookies, where 86% of NLEs have a penetration below 0.5% in the quarter directly after launch and again in the quarter one year after launch.

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Table 3 also shows the proportion of NLEs with a penetration below the average penetration in the category. Compared to the average penetration of all products in the category, an average three in four NLEs have a penetration below the average in its category in the quarter after launch (M= 74%, SD= 2). The proportion decreases slightly to two in three NLEs with a penetration below its category average in the third year after launch (M= 67%, SD= 6).

<INSERT FIGURE 2 AROUND HERE>

Repeat buyer rate

After a NLEs' initial launch period (where 100% of NLE buyers are new), around one in five buyers of the NLE had bought the NLE in its launch quarter (M= 19%, SD= 7). Table 4 shows the mean percentage of buyers who purchased the NLE previously for the 'average' NLE in the five categories. The mean composition of repeat NLE buyers steadily increases each quarter after launch. Almost 30% of NLEs' buyers had purchased the NLE during its first year in the market. Looking further, the buyer profile of the 'average' NLE is comprised of around two in three buyers who had purchased the NLE previously in the last two (M= 35%, SD=7) and three years (M= 39%, SD= 7) after introduction.

<INSERT TABLE 4 AROUND HERE>

Figure 3 is an example in the Coffee category of the distribution of how many repeat buyers each NLE has in the key quarters. The distribution patterns in all categories and quarters show NLE repeat buyer bases peaking around zero, 50% and 100%. Although it's possible some NLEs recruit all new buyers or have all past buyers purchase it again in the following

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quarters, it is likely the peaks are an artefact of NLEs that have very small customer bases (i.e., one or two people buying).

<INSERT FIGURE 3AROUND HERE>

Bought parent brand rate

NLEs are more often bought by category buyers who had purchased the parent brand at least once prior to launch. Table 5 shows an average nine in 10 buyers of NLEs had purchased the parent brand before launch (M= 90%, SD= 4). This rate was persistent in all the categories investigated.

<INSERT TABLE 5 AROUND HERE>

Figure 4 shows that there is variation in the percentage of NLE buyers that were past parent brand buyers. While Figure 4 is an example in the Cereal category, the distributions in all categories have a negative skew. The mean parent brand rate results show the average NLE does not attract new buyers to the parent brand and majority of NLEs have buyer bases were 90% of buyers had bought the parent brand previously.

Around two in three NLEs had buyer bases where 90% or more were previous parent buyers in the quarter directly after launch (M= 63%, SD= 10), which continues in the first, second and third year after launch. Interestingly, Cereal had the lowest percentage of NLEs with previous parent buyer bases above 90%, but this was still the largest NLE group (Figure 4).

<INSERT FIGURE 4 AROUND HERE>

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Parent brand sales share

NLEs bring in around 13% of the parent brand's sales revenue, on average in the quarter after launch (SD= 5), but also one, two and three years after launch (M= 13%, SD= 5). Table 6 shows the striking steadiness in parent brand revenue share over time.

<INSERT TABLE 6 AROUND HERE>

However, the mean contribution of NLE to parent brand sales was not representative of what the majority or typical NLEs achieve in each category. The distributions have a large range (see example of NLEs in the Air Freshener category in Figure 5) and most NLEs contribute less than 10% to parent sales. Across the categories, two in three NLEs that were introduced in the previous quarter contributed less than 10% to parent brand sales (M= 64%, SD= 13). The percentage of NLEs contributing less than 10% to parent sales is lower in Cereal (48%) but higher in Air Freshener (81%, shown in Figure 5). NLEs that had 100% parent brand share (i.e., only parent brand product bought in the quarter) were removed in this analysis.

<INSERT FIGURE 5 AROUND HERE>

Discussion

Around a third of NLEs fail in the first quarter after launch, 50% fail in the year after launch and the failure rate reaches the 80% mark that is widely believed in practice (see Castellion and Markham, 2013) but only three years after launch. The rates in the first year after launch are higher than a recent study that use a similar measurement approach (Victory et al., 2021), but this could be due to the sample investigated (i.e., primarily non-seasonal products

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introduced by new and established brands) and data used (i.e., retail scanner) in that study. However, the failure rates in this study halve in the first year after launch once the more 'temporary' NLEs were removed. These findings show context is important when reporting new product failure rates, including aspects like the new product sample, the data and measurement approach and the observation timeframe.

NLEs seemingly have low performance on the five key performance indicators, with mean market share and penetration below 0.5% in each quarter. Mean market share and penetration are stable after the first-year mark. The relatively low market share and penetration values demonstrate the limitation in relying on NLEs to grow brand penetration and market share. Should NLEs be part of the long-term strategy of the brand, this supports the belief of the importance of new products of gaining consumer acceptance in the first 52-weeks after introduction. While there is some stability in mean results, the aggregate average of what the 'average' NLE achieve hide the variation of how most the 'typical' NLE achieves. For example, the market share and the penetration NLEs achieve peak close to zero, and most have a performance below the average in the category, but the distributions have a long tail.

The bought parent brand rate was surprisingly stable. An average nine in ten buyers of NLEs had bought the parent brand in a previous period. The buyer profiles for NLEs do vary (negative skewed) but the majority of NLEs had buyer bases where 90% or more of buyers had bought the parent brand previously. This supports past research that showed buyers of new variants are skewed towards parent brand buyers (Tanusondjaja et al., 2016). The finding helps to show that a marketing goal of launching new products to capture a different target segment is extremely difficult: new items overwhelmingly appeal to people who already buy the parent brand.

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Adding to the NLE brand incrementality results, the 'average' NLE contributes around 13% to parent brand sales in each quarter after introduction. While some NLEs are top sellers for their parent brands, most NLEs represent less than a 10% share to its parent brand. This provides additional evidence that NLEs have limited efficacy in attracting new buyers to the parent brand and are also unlikely to be a major contribute to parent brand sales revenue, at least in the three years from launch. Instead most NLEs will join part of the 'long tail' of contribution to sales to its parent brand (see Tanusondjaja et al., 2018).

Implications

It is important for research in marketing to examine the conditions and moderators in empirically generalized relationships (Hanssens, 2018). Understanding the conditions when new products are more successful than others is a key area in NPD research (Lee and Markham, 2016) but this is not possible until academics benchmark how new products perform on key indicators (Victory et al., 2021). This study helps to build this foundation by documenting how NLEs perform on performance indicators that are commonly used in practice to evaluate success.

The results from this study conclude while most non-seasonal NLEs do not fail in the shorter term (i.e., 70% failure not reached until three years after launch), most NLEs have a market share and penetration that is just around or below the average in its category. Few of the NLEs investigated could be deemed truly successful in the marketplace. Furthermore, the introduction of a NLE tends not to be an effective strategy to attract new buyers to its parent brand, and most do not contribute more than 10% of sales to its parent brand. These findings are important to inform practitioner expectations about what to expect from NLEs.

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Our study provides practitioners involved in NPD activities a benchmark around how NLEs perform. However, we encourage practitioners to use the measures used in this paper to benchmark how NLEs perform in their categories, against their own launches. More studies are opting to observe how new products perform in consumer purchasing data, but much of the current benchmarks are based on surveys and subjective practitioner evaluation. This study confirms different success measurement approaches do yield different results depending on the context of the measurement. It is our hope that this study will encourage more academics to measure new product success using replicable measures, in practitioner relevant data.

This study's most substantial contribution is to marketing science, including informing *when* to measure new product success and identifying *how* new product success is distributed. First, we show mean market share and penetration steadies one year after launch, providing peer-reviewed evidence that a NLEs' first year is important in consumer acceptance. This finding is very useful in judging the likely future success of an NLE: these findings suggest that if a consumer goods NLE is performing poorly in the first year, it is unlikely to markedly improve its performance in the second and third year. Second, while mean success results are stable, the distributions reveal the mean results are misleading and do not reflect the spread in how NLEs perform. Our findings show the importance of understanding how performance is distributed to inform sampling decisions but also when making assumptions around which statistical analyses to use. This is because performance indicators we explored had a positive skew and were not equally distributed.

Limitations & Future Research

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Given the variation in how NLE perform around the mean rates, we encourage academics to examine how success measures relate together, to reveal when success can vary. For example, do 'successful' NLEs that remain in the market for longer also have greater sales performance as early as their launch? This would identify whether it is possible to 'pick' a winner at launch, based on the initial performance achieved by new product introductions.

This study investigated NLEs that were introduced from a range of established parent brands. However, it is not known how the success measures vary for NLEs introduced by parent brands with higher and lower market shares or shares that are growing or declining in share. The key question becomes, do certain brands have a higher chance of launching a successful NLE? There is evidence that new consumer products launched by higher share brands have fewer failures (Victory et al., 2021) and more trial and repurchase (Sinapuelas et al., 2015), compared to lower market share brands. However, it is not widely understood how the ability of NLEs to attract new brand buyers and the sales importance of NLEs varies by parent brand sales and dynamics. We encourage future research to examine how NLE success, particularly parent brand success measures, vary by parent brand market share and share dynamics.

Note

Researcher(s) own analyses calculated (or derived) based in part on data from The Nielsen Company (US), LLC and marketing databases provided through the Nielsen Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business.

The conclusions drawn from the Nielsen data are those of the researcher(s) and do not reflect the views of Nielsen. Nielsen is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein.

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Tables and Figures

Table 1 New line extension failure rate

	N	NEW LINE EXTENSION FAILURE RATE - %												
		L	Q1	Q2	Q3	Y1	Q5	Q6	Q7	Y2	Q9	Q10	Q11	Y3
COOKIES	17263	0	37	44	48	52	57	61	64	67	71	75	79	84
COFFEE	5998	0	31	36	40	43	47	51	54	57	61	65	69	75
CEREAL	5372	0	25	33	38	43	47	50	54	57	60	63	67	72
TOOTHPASTE	5039	0	34	44	51	56	61	64	67	69	72	75	78	83
AIR FRESHENER	3322	0	34	44	50	55	59	63	66	69	72	76	80	84
TOTAL/MEAN	**36994**	**0**	**32**	**40**	**45**	**50**	**54**	**58**	**61**	**64**	**67**	**71**	**75**	**80**
STD DEVIATION	**5603**	**0**	**5**	**5**	**6**	**6**	**7**	**7**	**6**	**6**	**6**	**6**	**6**	**6**

	N	NEW LINE EXTENSION FAILURE RATE (OMITTING Q1 FAILURES) - %												
		L	Q1	Q2	Q3	Y1	Q5	Q6	Q7	Y2	Q9	Q10	Q11	Y3
COOKIES	10856	0	0	11	17	20	28	38	43	47	54	60	66	74
COFFEE	4127	0	0	7	12	16	20	28	33	37	43	49	55	64
CEREAL	4037	0	0	10	18	22	26	33	38	42	46	51	56	62
TOOTHPASTE	3326	0	0	16	26	30	38	46	50	54	58	62	67	74
AIR FRESHENER	2200	0	0	16	25	26	31	45	49	53	58	64	69	77
TOTAL/MEAN	**24546**	**0**	**0**	**12**	**20**	**23**	**29**	**38**	**43**	**47**	**52**	**57**	**63**	**70**
STD DEVIATION	**3413**	**0**	**0**	**4**	**6**	**5**	**7**	**8**	**7**	**7**	**7**	**7**	**7**	**7**

Table 2 New line extension mean revenue market share

	N	NEW LINE EXTENSION MEAN REVENUE MARKET SHARE - %					
		Q1	Q2	Q3	Y1	Y2	Y3
COOKIES	17263	0.1	0.1	0.1	0.1	0.1	0.1
COFFEE	5998	0.2	0.2	0.2	0.2	0.2	0.2
CEREAL	5372	0.1	0.2	0.2	0.2	0.2	0.2
TOOTHPASTE	5039	0.3	0.3	0.3	0.3	0.3	0.3
AIR FRESHENER	3322	0.6	0.6	0.6	0.6	0.6	0.7
TOTAL/MEAN	**36994**	**0.3**	**0.3**	**0.3**	**0.3**	**0.3**	**0.3**
STD DEVIATION	**5603**	**0.2**	**0.2**	**0.2**	**0.2**	**0.2**	**0.2**

	N	NEW LINE EXTENSION REVENUE MARKET SHARE BELOW CATEGORY MEAN - %					
		Q1	Q2	Q3	Y1	Y2	Y3
COOKIES	17263	75	75	75	74	76	76
COFFEE	5998	71	69	68	66	66	66
CEREAL	5372	72	69	67	67	63	61
TOOTHPASTE	5039	75	73	70	71	73	71
AIR FRESHENER	3322	67	68	66	61	60	56
TOTAL/MEAN	**36994**	**72**	**71**	**69**	**68**	**68**	**66**
STD DEVIATION	**5603**	**3**	**3**	**4**	**5**	**7**	**8**

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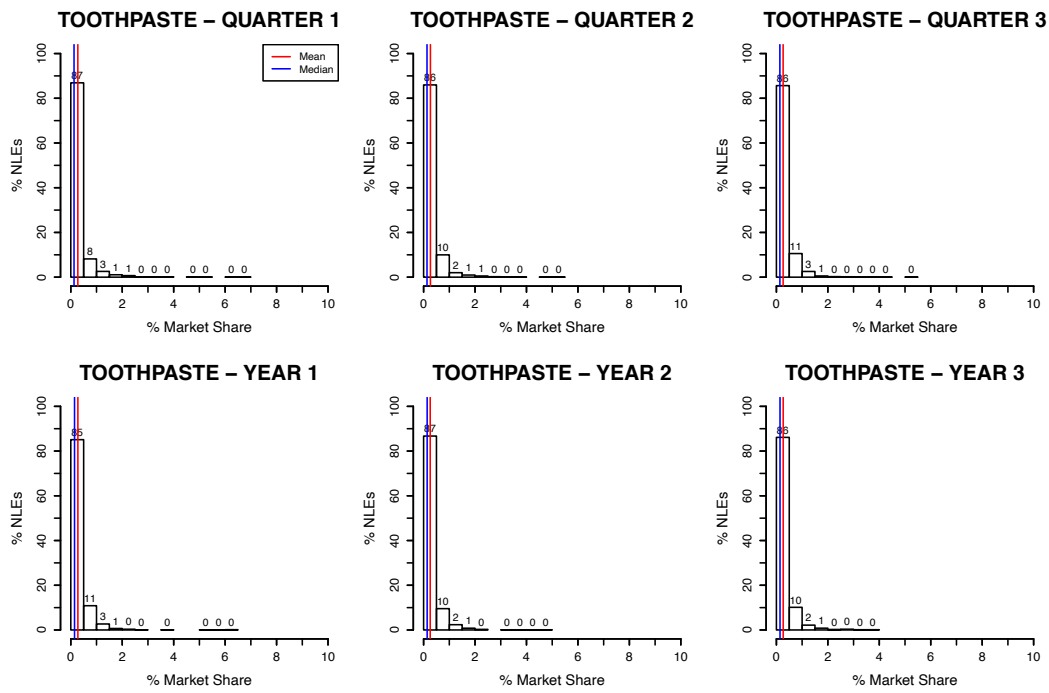


Figure 1 New line extension sales revenue market share distribution

Table 3 New line extension mean penetration

	N	NEW LINE EXTENSION MEAN PENETRATION - %					
		Q1	Q2	Q3	Y1	Y2	Y3
COOKIES	17263	0.3	0.3	0.3	0.3	0.3	0.3
COFFEE	5998	0.4	0.4	0.5	0.5	0.5	0.4
CEREAL	5372	0.6	0.6	0.7	0.7	0.8	0.8
TOOTHPASTE	5039	0.4	0.4	0.4	0.4	0.4	0.4
AIR FRESHENER	3322	0.8	0.9	0.9	1.0	0.9	1.1
TOTAL/MEAN	**36994**	**0.5**	**0.5**	**0.6**	**0.6**	**0.6**	**0.6**
STD DEVIATION	**5603**	**0.2**	**0.2**	**0.2**	**0.3**	**0.3**	**0.3**

	N	NEW LINE EXTENSION PENETRATION BELOW CATEGORY MEAN - %					
		Q1	Q2	Q3	Y1	Y2	Y3
COOKIES	17263	74	73	73	73	74	75
COFFEE	5998	75	72	70	68	67	67
CEREAL	5372	72	69	67	66	63	62
TOOTHPASTE	5039	77	76	74	71	71	70
AIR FRESHENER	3322	73	68	70	64	64	59
TOTAL/MEAN	**36994**	**74**	**72**	**71**	**68**	**68**	**67**
STD DEVIATION	**5603**	**2**	**3**	**3**	**4**	**5**	**6**

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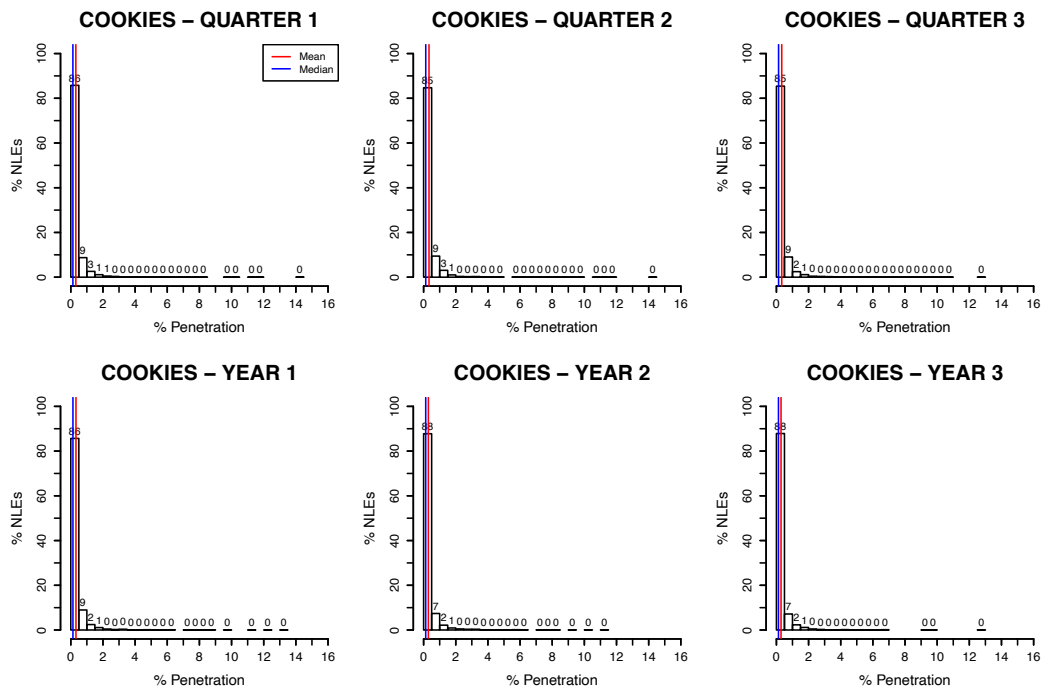


Figure 2 New line extension penetration distribution

Table 4 New line extension mean repeat buyer rate

	N	NEW LINE EXTENSION MEAN REPEAT BUYER RATE - %					
		Q1	Q2	Q3	Y1	Y2	Y3
COOKIES	17263	23	25	29	26	29	32
COFFEE	5998	28	32	35	36	44	47
CEREAL	5372	20	27	33	33	42	44
TOOTHPASTE	5039	12	18	22	22	32	35
AIR FRESHENER	3322	13	18	19	22	28	35
TOTAL/MEAN	**36994**	**19**	**24**	**28**	**28**	**35**	**39**
STD DEVIATION	**5603**	**7**	**6**	**7**	**6**	**7**	**7**

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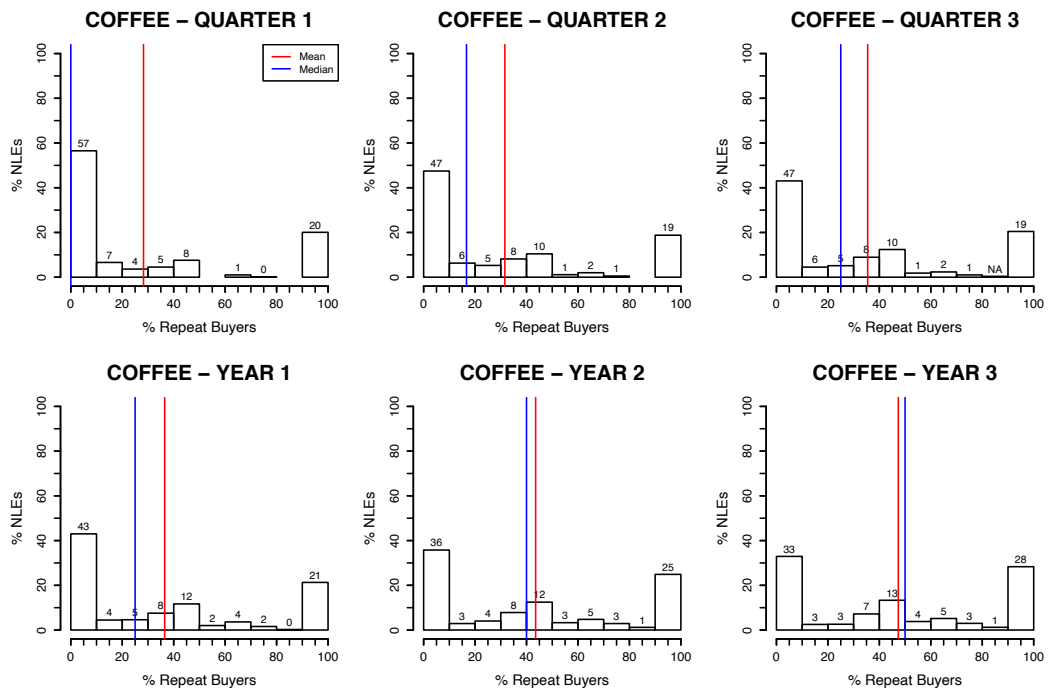


Figure 3 New line extension repeat buyer rate distribution

Table 5 New line extension bought parent brand rate

	N	NEW LINE EXTENSION BOUGHT PARENT BRAND RATE - %					
		Q1	Q2	Q3	Y1	Y2	Y3
COOKIES	14310	87	86	86	87	88	88
COFFEE	5327	90	90	90	90	91	91
CEREAL	4831	82	81	81	81	80	79
TOOTHPASTE	4851	91	91	90	90	89	90
AIR FRESHENER	3000	94	93	93	93	93	92
TOTAL/MEAN	**32319**	**89**	**88**	**88**	**88**	**88**	**88**
STD DEVIATION	**4475**	**5**	**5**	**5**	**5**	**5**	**5**

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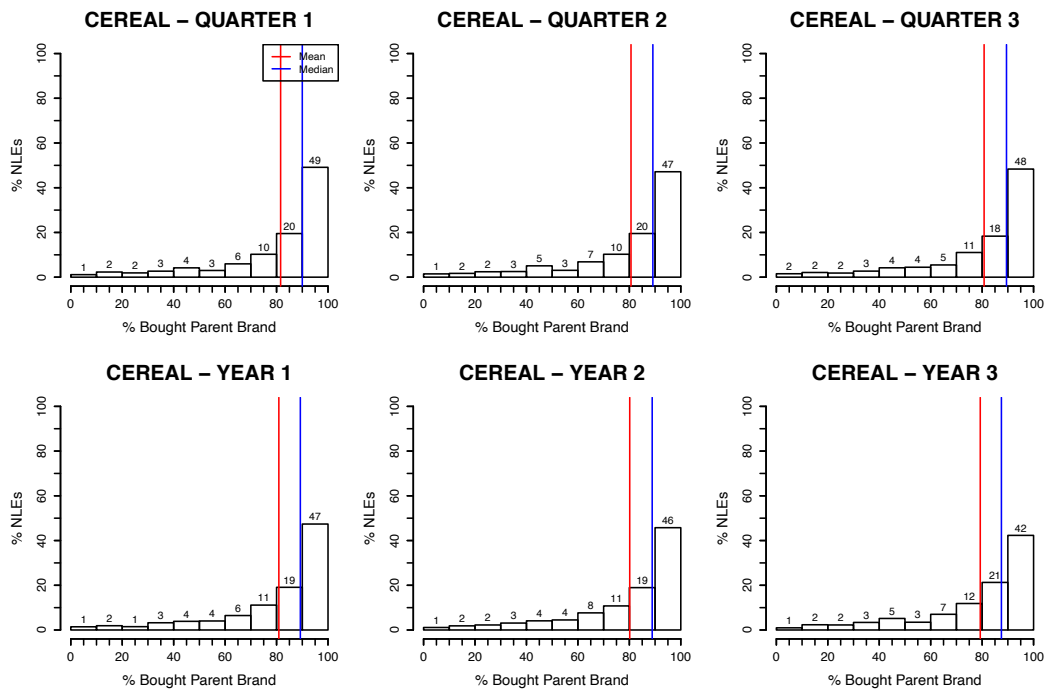


Figure 4 New line extension bought parent brand rate distribution

Table 6 New line extension mean parent brand revenue sales share

	N	NEW LINE EXTENSION PARENT BRAND REVENUE SALES SHARE - %					
		Q1	Q2	Q3	Y1	Y2	Y3
COOKIES	14310	16	16	16	15	15	14
COFFEE	5327	13	13	12	12	11	11
CEREAL	4831	20	21	21	21	22	22
TOOTHPASTE	4851	10	10	10	11	11	11
AIR FRESHENER	3000	7	8	8	8	8	9
TOTAL/MEAN	**32319**	**13**	**14**	**13**	**13**	**13**	**13**
STD DEVIATION	**4475**	**5**	**5**	**5**	**5**	**5**	**5**

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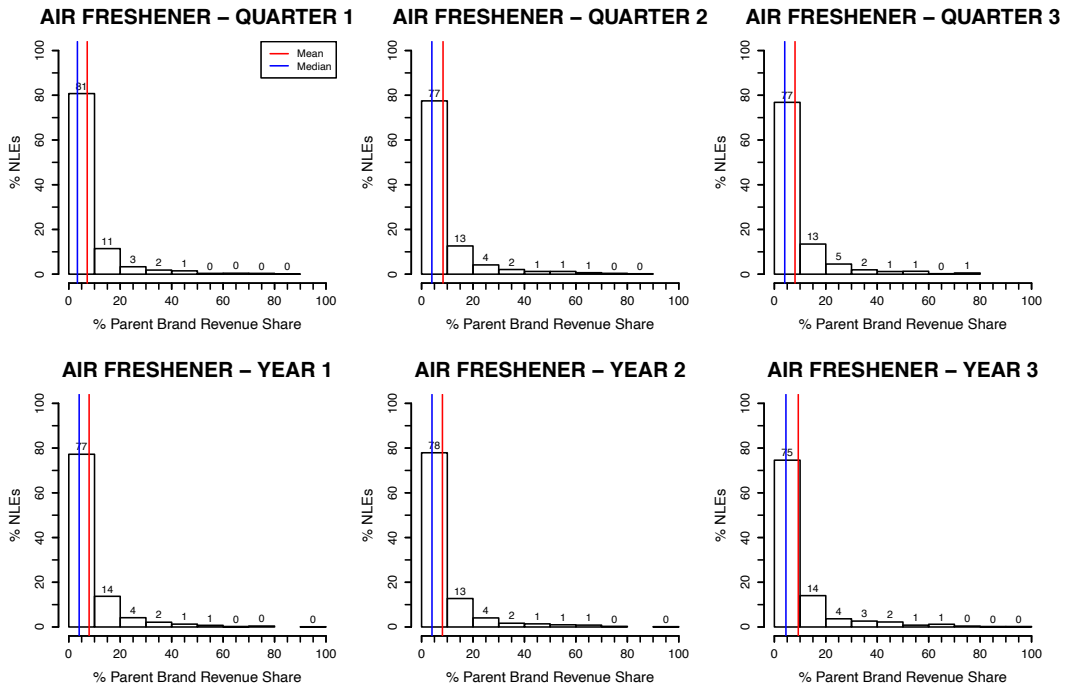


Figure 5 New line extension parent brand revenue sales share distribution

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